BOREHOLE SITING REPORT FOR BOITSHWARELO JOSEPH

DATE: OCTOBER 2025

INTRODUCTION

This borehole survey report is the outcome of a survey conducted at Mrs Boitshwarelo Joseph's property at Mosa Lands in October 2025. The objective of the survey was to establish potential borehole fractures capable to provide sustainable ground water for Agricultural and domestic use. This report details the survey activities, results, and recommendation.

SURVEY EQUIPMENT AND METHOD

Audio Magneto telluric (AMT)

An Audio Magneto telluric (AMT) receiver S 500 from PQWT with the ability to site as far as 500m depth was used in this survey. The receiver creates 2D sections of the earth's apparent resistivity sections. In different geological formations, dykes , fractures and faults can be precisely located and these are good location of siting a water borehole.

The magnetotelluric method measures the earth's electric and magnetic fields and this is used to infer electrical resistivity structure of the subsurface. The method has great depth of penetration and can provide information in poor seismic and non-seismic areas and on poor and good conductors.

The receiver has got 2 electrodes used in a profile line of 10m apart and with 1 meter station spacing. Survey Profiles are produced showing apparent resistivity variation in both vertical and horizontal directions. Faults, fractures and dykes are then easily identified.

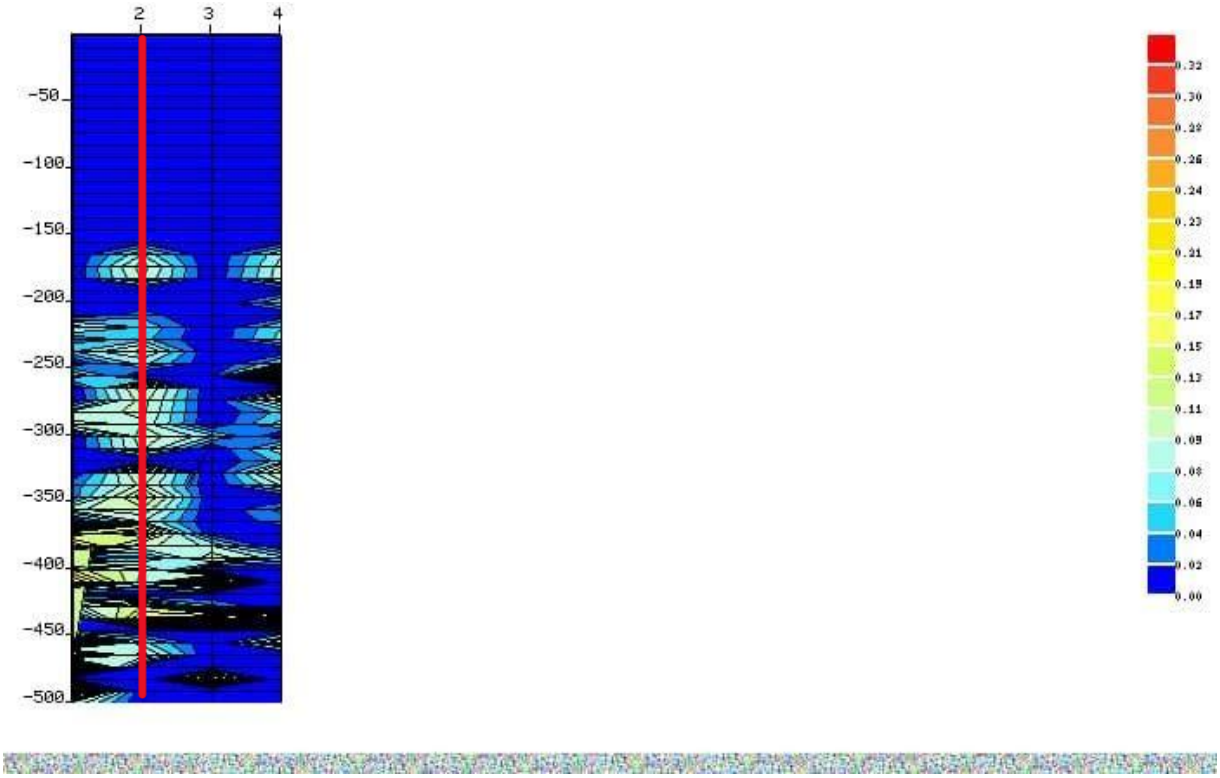
SURVEY

An appropriate geophysical traverse line layout plays an important role in the collection and analysis of subsurface geophysical data that accurately represent site properties. 1 geophysical traverse line was designed.

PROFILE LINES

A survey lines was profiled using a 1m station spacing. A traverse line showed potential water bearing fractures

The AMT Profile Line 1(Priority)



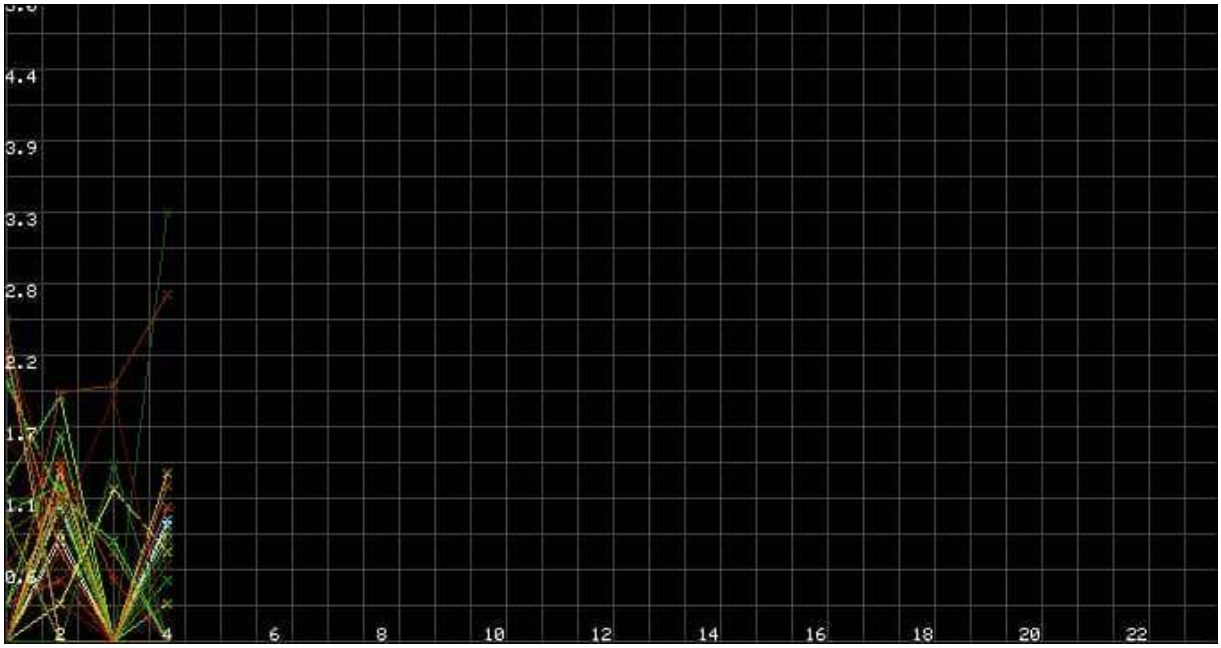


Figure 1: Traverse line

RESULTS

THE SITE CHOSEN AS PRIORITY IS AT POINT 4 ON THE PROFILE LINE AND CAN BE DRILLED TO THE MAXIMUM DEPTH OF 340M WITH POTENTIAL WATER STRIKES FROM 190M,250M AND 280M. A PEG WAS INSERTED AS A MARKER AND WAS SHOWN TO THE CLIENT.

THE BOREHOLE SURVEY WAS DONE IN GOOD FAITH.

RECOMMENDATIONS

- Drill rig to position on the identified point
- Borehole conditioning: The driller or supervisor must ensure that after drilling to the required depth, especially with an air rig, they must wait for about 1 hour before blowing water out to clean the borehole. This ensures that all the mud cake produced during drilling will be removed from the borehole walls. This will cause all fractures to be open and therefore more recharge to the borehole.
- Test pumping required to measure the exact yield of borehole for future supply and borehole management.

REPORTED BY ADOSA GROUNWATER LOCATORS (PTY) Ltd

O Marumo, K Marumo & B Botsang (Surveyors)

Ado-sa Groundwater Locater (Pty) Ltd
P. O. Box 26 ABG, Gaborone
Tel: (+267) 73830945 / 71851794
Email: adosagw@gmail.com